PYTHON MINI PROJECT

Title: Personal Expense Tracker using File Integration

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AIM

To design and develop a Personal Expense Tracker using Python that reads and writes data to the file system for storing daily expenses.

OBJECTIVE

- To understand file integration in Python
- To perform read and write operations on files
- To manage real-life expense data
- To apply functions and data structures
- To generate reports from stored data
- To build a menu-driven application

PROCEDURE (Detailed)

1. Open Python IDLE or any editor and create a file named expense_tracker.py.
2. Create a text file named expenses.txt to store expense records.
3. Define a function add_expense():
   - Accept date, category, and amount
   - Write the data into the file
4. Define a function view_expenses():
   - Read all records from the file

    - Display them properly
5. Define a function total_expense():
    - Read file data
    - Calculate total expenses
6. Define a function search_expense():
    - Search expenses by category
7. Use a menu-driven system:
    - Add Expense
    - View Expenses
    - Total Expense
    - Search Expense
    - Exit
8. Use a loop to allow repeated operations.
9. Handle file not found cases.
10. Ensure proper opening and closing of files.
11. Test with multiple entries.

PYTHON CODE

```
import os

FILE_NAME = "expenses.txt"


def add_expense():
    date = input("Enter Date: ")
    category = input("Enter Category: ")
    amount = input("Enter Amount: ")

    with open(FILE_NAME, "a") as file:
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        file.write(date + "," + category + "," + amount + "\n")

    print(" Expense Added!")


def view_expenses():
    if not os.path.exists(FILE_NAME):
        print("No records found!")
        return

    print("\n--- Expense Records ---")
    with open(FILE_NAME, "r") as file:
        for line in file:
            date, category, amount = line.strip().split(",")
            print("Date:", date, "| Category:", category, "| Amount:", amount)


def total_expense():
    total = 0

    if not os.path.exists(FILE_NAME):
        print("No records found!")
        return

    with open(FILE_NAME, "r") as file:
        for line in file:
            _, _, amount = line.strip().split(",")
            total += float(amount)

    print("x Total Expense:", total)


def search_expense():
    search = input("Enter category to search: ")

    if not os.path.exists(FILE_NAME):
        print("No records found!")
        return
```

```
    found = False
    with open(FILE_NAME, "r") as file:
        for line in file:
            date, category, amount = line.strip().split(",")
            if category.lower() == search.lower():
                print("Date:", date, "| Category:", category, "| Amount:", amount)
                found = True

    if not found:
        print("No matching records!")


def main():
    while True:
        print("\n--- Expense Tracker ---")
        print("1. Add Expense")
        print("2. View Expenses")
        print("3. Total Expense")
        print("4. Search Expense")
        print("5. Exit")

        choice = int(input("Enter choice: "))

        if choice == 1:
            add_expense()
        elif choice == 2:
            view_expenses()
        elif choice == 3:
            total_expense()
        elif choice == 4:
            search_expense()
        elif choice == 5:
            print("Exiting...")
            break
        else:
            print("Invalid choice!")
```

main()

OUTPUT

```
--- Expense Tracker ---
1. Add Expense
2. View Expenses
3. Total Expense
4. Search Expense
5. Exit

Enter choice: 1
Enter Date: 10-03-2026
Enter Category: Food
Enter Amount: 200
Expense Added!

Enter choice: 3
Total Expense: 200
```

RESULT

The Personal Expense Tracker was successfully implemented using file integration. The program stored and retrieved expense data efficiently from a text file.

CONCLUSION

Thus, the mini project was successfully developed using Python. It demonstrated practical usage of file handling, data storage, and real-life financial tracking applications.